

Final Report: MSc Cybersecurity Admission Project

[Your Name]

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GitHub Repository

<https://github.com/jafarizadeh/cybersup.git>

Executive Summary

This report documents the three practical exercises completed as part of the MSc Cybersecurity admission process at Cybersup. The exercises cover core competencies in:

- **Networking:** VLAN segmentation, DHCP, and inter-VLAN routing on Cisco devices.
- **System Administration:** Active Directory deployment and user/group automation via PowerShell.
- **DevOps/Containerization:** End-to-end WordPress deployment using Docker Compose.

All configurations, scripts, and supporting screenshots are available in the GitHub repository above.

Project Overview

Exercise	Domain	Tools & Technologies	Deliverables
1	Networking	Cisco Packet Tracer, Cisco IOS	.pkt file, device configs, README
2	System Admin	Windows Server 2019/2022, PowerShell, CSV	PowerShell scripts, CSV, README
3	DevOps	Docker, Docker Compose, Nginx, PHP-FPM, MariaDB, WordPress	docker-compose.yaml, nginx.conf, README

1 Exercise 1 – Networking with Cisco Packet Tracer

1.1 Objectives

- Design a small-scale enterprise network with four VLANs.
- Configure DHCP services on a Cisco 1941 router.
- Enable inter-VLAN routing and prepare for Internet connectivity.

1.2 Equipment & Topology

- **Router:** Cisco 1941
- **Switches:** 3 × Cisco (simulated in Packet Tracer)
- **Access Points:** 3 × PT-AC
- **Clients:** 6 desktop PCs, 3 laptops
- **IP Phones:** 3 × Cisco 7960

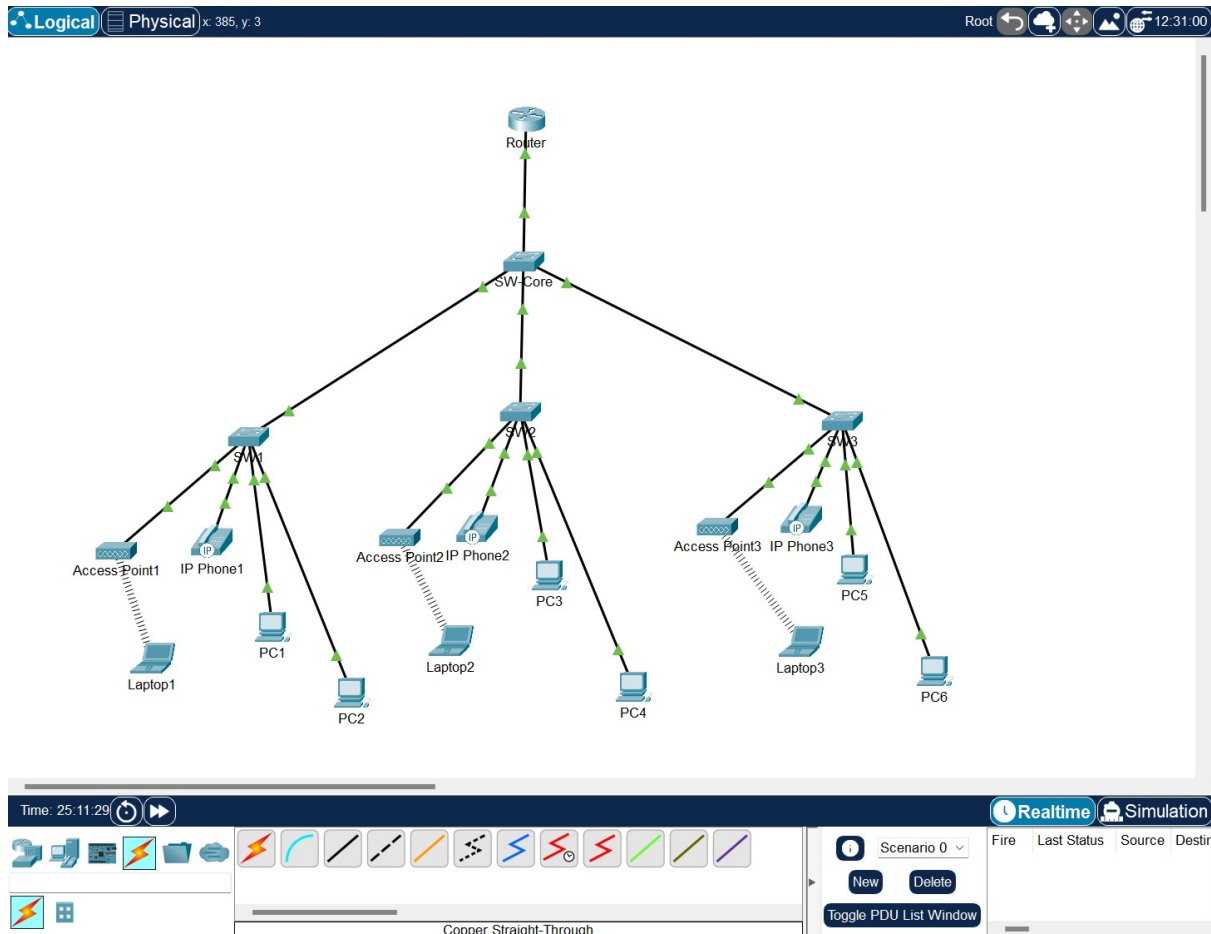


Figure 1: Logical Network Topology

Logical Topology:

1.3 VLAN & DHCP Configuration

VLAN ID	Name	Subnet	DHCP Range	Gateway
1	VoIP	192.168.0.0/24	.10-.50	192.168.0.1
10	WiFi	192.168.10.0/24	.10-.50	192.168.10.1
20	PC	192.168.20.0/24	.10-.50	192.168.20.1
30	Admin	192.168.30.0/24	.10-.50	192.168.30.1

1.4 Implementation Steps

1. Define VLANs and assign access/trunk ports on each switch.
2. Configure sub-interfaces on the router with `encapsulation dot1Q` and DHCP pools.
3. Connect devices, power on, and verify DHCP address allocation.
4. Test inter-VLAN connectivity using ping.

1.5 Validation & Results

- **Success:** All clients obtained IP addresses via DHCP.
- **Success:** Inter-VLAN routing validated by cross-VLAN pings.
- **Success:** Router reachability confirmed from each subnet.

2 Exercise 2 – Active Directory Automation with PowerShell

2.1 Objectives

- Deploy an Active Directory domain `laplateforme.io` on Windows Server.
- Automate creation of 15 users and 9 groups from a CSV file.
- Enforce a strong password policy and require first-login password change.

2.2 Environment Preparation

- OS: Windows Server 2019/2022
- Static IP (e.g., 192.168.1.100)
- Active Directory Domain Services (AD DS) role installed

2.3 CSV Import Script

Sample CSV:

```
nom,prénom,groupe1,groupe2,...
ALEXANDRE,MARCELLINE,Animation,
ARAGON,ISABELLE,Animation,
...
```

Scripts Provided:

- `AD-Setup.ps1`: Installs AD DS role and promotes server.
- `Create-UsersFromCSV.ps1`: Reads CSV, creates groups, users with default password `Azerty2025!`, enforces password change, and assigns groups.

2.4 Password & Security Policies

```
Set-ADDefaultDomainPasswordPolicy ‘  
-Identity laplateforme.io ‘  
-ComplexityEnabled $true ‘  
-MinPasswordLength 8 ‘  
-MaxPasswordAge 42.00:00:00 ‘  
-LockoutThreshold 5 ‘  
-ReversibleEncryptionEnabled $false
```

2.5 Implementation Steps

1. Rename server and assign static IP.
2. Install AD DS role and reboot.
3. Promote to domain controller using `Install-ADDSForest`.
4. Run `Create-UsersFromCSV.ps1` to import users and groups.
5. Verify all accounts and memberships in Active Directory Users and Computers.

2.6 Validation & Results

- Domain and DNS configured successfully.
- All 15 users and 9 groups created and correctly assigned.
- Password policy enforced and first-login password change required.

3 Exercise 3 – Containerized WordPress with Docker Compose

3.1 Objectives

- Deploy a scalable WordPress site using Docker Compose.
- Use persistent volumes to retain data across container restarts.

3.2 Service Architecture

- **mariadb**: Database service (volume `db_data`)
- **php-fpm**: PHP processor (volume `wordpress_data`)
- **wordpress**: CMS application
- **nginx**: Web server, port mapping 8080:80

3.3 Configuration Files

`docker-compose.yaml` and `nginx.conf` define services, volumes, networks, and Nginx settings (root, PHP upstream, static caching).

3.4 Deployment Steps

1. Clone repository and navigate to `/exercise3/`.
2. Run `docker-compose up -d`.
3. Verify containers are running (`docker ps`).
4. Access `http://localhost:8080` and complete WordPress setup.

3.5 Validation & Results

- All containers started without errors.
- Data persisted after container restarts.
- WordPress installation wizard accessible and functional.

Challenges & Solutions

Challenge	Solution
VLAN misconfiguration on trunk ports	Reviewed and corrected <code>switchport trunk native vlan</code> settings
PowerShell errors during group creation	Added Try/Catch blocks with logging
Docker volume permission issues	Adjusted host folder ownership and permissions

Conclusion

The admission project successfully demonstrates:

- **Network Engineering:** VLAN design, DHCP, inter-VLAN routing on Cisco.
- **Systems Administration:** Automated AD deployment and user management via PowerShell.
- **DevOps:** Container orchestration with Docker Compose and Nginx.

For detailed configurations, scripts, and screenshots, please explore the subdirectories:

- `/exercise1/`
- `/exercise2/`
- `/exercise3/`